

Star Visibility Calculator During Twilight

A clear, organized explanation of every input field, option, and result column - written for readers who are not specialists.

<https://starsvisibility.pages.dev/>

What does this guide include?

It explains what each field means, which default is normally appropriate, and when there is a good reason to change it.

What is not included?

This is not a halachic ruling, and it does not claim that the calculator's output is certain without local sky measurements.

Recommended basic use

Select a location and date, choose Evening mode, include the Moon, use the Physical model, and select the sky quality that matches actual conditions.

Core reading rule

First visible is a threshold moment. To judge comfortable visibility, also check **Best-night status** and **Confidence**.

Contents and Quick Start

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Basic use: Most calculations do not require advanced data. Select a location, date, Evening mode, an appropriate sky quality, include the Moon, and use the Physical model.

1 Select a location

Begin with Location Search. If the location appears in the list, there is no need to enter coordinates manually.

2 Date and mode

Select the local date. For normal use at the beginning of night, choose **Evening**.

3 Basic defaults

Minimum altitude 3°, magnitude limit 6.5, 2-minute resolution, and the Physical model.

4 Sky conditions

Choose Dark site, Clear horizon, Urban sky, or Haze/Dust. Enter SQM or kV only when supported.

5 Read the result

Focus on First visible, Sun altitude, Mag, Star altitude, Limiting factor, Confidence, and Best-night status.

A. Basic Concepts Before You Begin

Sun altitude

When the Sun is below the horizon, the value is negative. -6° is strong twilight; -12° is darker; about -18° marks the end of astronomical twilight.

Star altitude

0° is the horizon, 10° is low, 45° is midway up the sky, and 90° is overhead. Low stars are harder to see.

SQM

A measure of sky darkness. A higher number means a darker sky. 21.8 is very dark; 18.8 is much more urban.

Magnitude

A measure of a star's brightness. The scale is reversed: a smaller number is brighter. Magnitude 1 is brighter than magnitude 5.

Confidence

The reliability level of the result. It does not say whether the star is visible; it indicates how dependent the calculation is on uncertain local conditions.

kV / extinction

How strongly the atmosphere weakens starlight in the V band. The value is larger in haze and for lines of sight close to the horizon.

B. Screen Layout

Overall structure of the main screen

Selection and Input Fields

Search location

Local date

Evening

Before sunrise

If it were rising

Sky quality / horizon twilight

Calculation model

Calculate

Results Table

Visible objects found

Earliest appearance

Catalog objects

First visible	Sun altitude	Mag	Star altitude	Confidence
20:07	-7.8°	1.2	16°	Medium

Advanced Options - Mainly for Measurements

kV

The atmospheric extinction coefficient. One of the most important inputs for low-altitude stars.

SQM / Bortle

The source used for background sky brightness. Enter only a measurement or a well-founded estimate.

Observer data

Age, pupil size, visual acuity, and dark adaptation when tailoring the estimate to a particular person.

Horizon obstruction

Buildings, trees, mountains, or another physical obstruction, entered in degrees.

Reading a Result Row

First visible	The moment the star crosses the visibility threshold.
Sun altitude	How far below the horizon the Sun is at that moment.
Star altitude	The star's height above the horizon.
Limiting factor	The main factor restricting visibility.
Confidence	The reliability of the calculation under the selected conditions.

First visible is a threshold moment and may be marginal. For a better sense of comfortable visibility, also check **Best-night status** and **Best visibility margin**.

C. Location and Time Fields

Field	Meaning	Recommended use
Location Search	Select a city or observing point from the list. The selection updates latitude, longitude, elevation, and time zone when available in the database.	The preferred way to begin a normal calculation. If the site is not listed, enter accurate coordinates manually.
Latitude / Longitude	The geographic coordinates of the observing site. They determine the altitudes of the Sun and stars relative to the local horizon.	Do not change them after selecting a listed location unless you need a more exact observing point.
Local date	The civil date at the observing site for which the calculation is performed.	Make sure this is the local date at the observing site.
IANA time zone / UTC offset	The time zone, such as Asia/Jerusalem or America/New_York. The UTC offset is calculated for the selected date and shown for reference.	Usually there is no need to edit this manually. It is filled automatically when the location is selected from the list.
Clock display	Only the time format: 24-hour display or AM/PM.	This does not alter the calculation; it changes only how times are displayed.

D. Calculation Mode

Option	What the calculator does	Recommended use
Evening	Scans after the beginning of night to find the first time the star crosses the visibility threshold.	The main mode for calculating the appearance of stars at the beginning of the evening.
Before sunrise	Scans before morning to find the last time the star remains visible before the background brightens.	Use this to examine the disappearance of stars in morning light.
If it were rising	A hypothetical mode: the star is placed at a fixed altitude and fixed angular distance from the Sun, and the calculator finds the required Sun depression.	Use for comparing visibility thresholds and stellar brightness, not for describing a real observing event.

E. Altitude, Magnitude, and Resolution

Field	Meaning	Recommended use
Minimum star altitude	The lowest apparent altitude at which a star may be counted in the result. The default is 3°.	The default suits an open horizon. For a blocked horizon, low haze, or a more conservative result, increase it to 5° or 8°.
Altitude in “If it were rising” mode	The hypothetical altitude assigned to the star in the comparison mode. The default is 10°.	Relevant only to “If it were rising.” It has no practical effect in Evening or Before sunrise mode.
Distance from the Sun in “If it were rising” mode	The hypothetical angular separation between the star and the Sun. The default is 90°.	Relevant only to the hypothetical mode. 30° is close to the Sun, 90° is intermediate, and 180° is opposite the Sun.
Magnitude limit	The depth of the catalog by magnitude. A larger magnitude number represents a fainter star. The default is 6.5.	6.5 is the normal naked-eye choice. 7.0 or 7.5 is for extended or professional investigation, not basic use.
Resolution in minutes	The initial time step of the scan. After a transition is found, the calculator refines the boundary.	The 2-minute default is suitable almost always. A result shown to the second is internal computational precision, not a guarantee of observational accuracy.

F. Sky Quality / Horizon Twilight

Professional note: Selecting a sky-condition preset is an estimate. A real SQM measurement and a measured or professionally estimated kV are always preferable to a general preset.

Option	Meaning	Recommended use
Dark site	Very dark sky, far from significant lighting, with an open horizon and good transparency.	Use for a genuinely dark observing site, such as a desert or mountain area, when there is no significant haze.
Clear horizon	Intermediate conditions: an open horizon and relatively clean sky, but not necessarily a professional dark site.	Use for an open location outside a city center when the background is not very bright and the horizon appears clear.
Urban sky	A bright sky background caused by city lighting or proximity to artificial light.	Use inside a city or in an area clearly affected by light pollution.
Haze / Dust	The atmosphere itself is murky: dust, heavy humidity, haze, smoke, or poor visibility.	Use when the horizon or sky appears milky or hazy. This affects both the background and the weakening of the star.

G. Moon, Calculation Model, and Probability Diagnostic

Field / option	Meaning	Recommended use
Moon effect - Include Moon	Moonlight is added to the sky background and reduces the limiting magnitude for faint stars.	Recommended for describing real conditions whenever the Moon is present and has an effect.
Moon effect - Ignore Moon	The calculation is performed as though the Moon contributes no light.	For comparison only: what the result would be without the Moon.
Physical model - Sky brightness	The main model. It calculates local background brightness, twilight, Moon, extinction, and the visibility threshold.	Recommended for normal calculations.
Corrected empirical model	A comparison model based on an empirical relationship between stellar brightness and Sun depression, with corrections.	For comparison and checking, not as the first choice for a general user.
Probability diagnostic from margin	A descriptive addition estimating how far the star is from the threshold. It does not change the appearance time.	Enable it when you want to know whether a result is marginal or comfortably beyond the threshold.

H. Advanced Sky Options

Important rule: Do not enter an exact number without a source. When no measurement is available, leave the field blank and use the default associated with the selected sky condition.

Field	Meaning	Recommended use
Zenith SQM	A measurement of sky darkness overhead in mag/arcsec ² . A higher value means a darker sky.	Enter only a real measurement or well-supported value. Otherwise leave it blank.
Bortle	A general sky-quality scale from 1 to 9: 1 is extremely dark and 9 is strongly urban.	Useful when SQM is unavailable but the site's class is reasonably known.
Measured SQM toward the star	A directional background measurement in the part of the sky where the star is located, rather than at the zenith.	Useful near the horizon, near a city, or near a light source. Enter it only when the measurement represents the relevant direction.
How directional SQM is applied	Determines whether the directional SQM value is applied to every row or not applied.	Apply it to all rows only when the measurement truly represents all directions being tested. Otherwise do not force it onto every row.
Low-altitude sky brightening	A manual correction for a brighter background low above the horizon.	Use only when the horizon is known to be brighter than the zenith and no directional SQM map is available. Enter physical blockage under Horizon obstruction.
Precision mode	Practical mode fills missing data; strict measurement mode emphasizes absent measurements and lowers confidence.	Use Practical mode for normal work. Use Strict measurement mode for professional review or reliability testing.
Directional / all-sky SQM map	A list of SQM measurements by azimuth and altitude, used to calculate the background in the star's direction.	For observers who have measured several directions and want the calculation to reflect differences across the sky.
Extinction coefficient kV	The V-band coefficient describing how strongly the atmosphere weakens starlight. Low means clear air; high means haze, dust, or humidity.	A very important input. Enter only a measurement or professional estimate; otherwise retain the default for the chosen sky condition.
Horizon obstruction	The altitude of a physical obstruction, in degrees, caused by buildings, trees, mountains, and so forth.	Use 0° for an open horizon; use a larger value when a real obstruction is known in the observing direction.
Temperature / air pressure	Inputs affecting atmospheric refraction, especially near the horizon.	Leave blank when unmeasured; their effect is usually smaller than the effect of kV or SQM.
Observer elevation above sea level	The altitude of the observing site in meters above sea level, used for horizon and refraction calculations.	Normally filled from the location list. This is not the observer's eye height above the ground.

I. Observer Data

Field	Meaning	Recommended use
Observer age	A parameter affecting the estimated sensitivity of the eye under dark conditions.	Enter only when tailoring the calculation to a particular observer. Otherwise leave blank.

I. Observer Data - Continued

Field	Meaning	Recommended use
Pupil diameter	Pupil diameter in millimeters. A larger pupil gathers more light and can allow fainter stars to be seen.	Enter only if known or when testing a defined scenario, such as 7 mm for a well dark-adapted eye or 5 mm for a more conservative case.
Visual acuity	A relative factor for visual sharpness. A value of 1 approximately represents normal vision.	Leave blank when no accurate value is available.
Dark adaptation	A value from 0 to 1: 0 means no adaptation and 1 means full adaptation.	Reduce it if the observer has just left a bright environment or has not spent enough time in darkness. Leave blank if unknown.

J. Star Catalog and Planets

Field	Meaning	Recommended use
Catalog depth	Controls how many faint stars are included: standard to 6.5, extended to 7.0, expert/experimental to 7.5.	For basic use choose 6.5. Higher limits are intended for borderline tests and greatly increase the number of rows.
Planets	Adds Mercury, Venus, Mars, Jupiter, Saturn, and Uranus to the calculation.	Enable when you want planets included; disable when you want fixed stars only.
Include planets	The checkbox that turns planetary inclusion on or off.	Clear the checkbox for fixed stars only.
Optional CSV catalog	Allows an external star catalog to be imported.	For advanced users. When no CSV is provided, the calculator uses its built-in catalog.

K. Summary Above the Table

Item	Meaning	Reading note
Visible objects found	The number of stars or planets that passed the visibility conditions under the selected settings.	A low number may result from the Moon, haze, light pollution, a high altitude cutoff, or low stars.
Earliest appearance	The object that first reached the visibility threshold in the evening, or the last object remaining visible before morning.	It is not necessarily the brightest object; it is the first one that satisfies all selected conditions.
Catalog objects	The total number of objects examined.	The number depends on catalog depth and whether planets are included.

L. Main Result Columns

Main columns: First visible, Sun altitude, Mag, Star altitude, Limiting factor, Confidence, and Best-night status.

Column	Meaning	Reading note
First visible	The time when the star first crosses the visibility threshold. Before sunrise: the last time it is still visible.	This is inherently a marginal moment; do not assume the star is already easy to see.
Sun altitude	The Sun's altitude at that moment. A negative number means the Sun is below the horizon.	A more negative value generally means a darker background.
Mag	The star's catalog magnitude.	A smaller number means a brighter star.
Brightness / appearance	A verbal description of the star's brightness or appearance.	Helps readers who are not accustomed to reading magnitude values.
Star / planet	The name of the star or planet.	Planets are marked separately from fixed stars.
Star altitude	The star's altitude above the horizon at the result time.	Low altitude reduces reliability and increases extinction. About 15° or higher is generally more stable than 3°-8°.
Limiting factor	The most important factor restricting visibility.	It may be twilight, Moon, light pollution, extinction, low altitude, or a combination.
Confidence	The confidence level for that row.	It does not decide visible/not visible; it indicates sensitivity to missing data or local conditions.
First-visible status	The visibility status at the exact first-appearance moment.	It will usually be marginal because this is the transition point.
Best-night status	The visibility status at the best time found during the scan.	Important for understanding whether the star becomes comfortably visible or only barely visible.

M. Advanced Result Columns - By Group

Advanced columns are intended for professional review and for understanding how a result was produced. They do not all need to be examined in basic use.

Color and Catalog

Column	Meaning
B-V	The star's color index: lower values tend toward blue; higher values tend toward red.
Color corr	A small correction added or subtracted because of the star's color.
Purkinje eye term	The part of the color correction associated with changes in the eye's sensitivity under dark conditions.
2nd-order color extinction	A small correction for differences in atmospheric extinction between colors.
Original mag / Used mag	The original magnitude compared with the magnitude actually used in the calculation.
VAR / Variability range / Variable type	Marks a variable star and gives its range and type when known.
DBL / Component mag / Combined mag	Data for a double or multiple star. Combined mag is the total light of the system when available.
Catalog source / uncertainty / notes	The source of the data and catalog warnings that affect the reliability of the row.

Position and Geometry

Column	Meaning
RA h / Dec deg	Astronomical coordinates of the star. Usually unnecessary for basic use.
Air mass	How much atmosphere the starlight passes through. A high value indicates a lower star and greater attenuation.
Sun az / Star az	Compass direction of the Sun and star: 0° north, 90° east, 180° south, 270° west.
Sun-star sep / Az diff / Sky side	The angular distance and directional relationship between the star and Sun. Proximity to the Sun brightens the background and makes visibility harder.

Physical and Empirical Models

Column	Meaning
Required depression (emp)	The required Sun depression according to the empirical model.
Required Sun alt (phys)	In the hypothetical mode, the required Sun altitude according to the Physical model.
Physical effective mag	The star's magnitude after extinction and color corrections in the Physical model.
Empirical final mag	The magnitude used by the empirical model, including the near-Sun penalty.
Phys visible / Emp visible	Whether the star is visible according to each of the two models.
Physical time / Empirical time	The appearance time according to each model.
Δ phys/emp min / Diff phys/emp	The time difference between the models and whether the gap is significant.
Near-Sun penalty (emp)	The empirical-model penalty for a star located close to the Sun.

Sky Brightness and Visibility Limit

Column	Meaning
Sky bright mag/arcsec ²	The sky-background brightness around the star. A higher value means a darker sky.
Local limit mag	The faintest magnitude the eye is estimated to detect at that moment and in that part of the sky.
First-visible margin	The margin above the threshold at first appearance. It is generally close to zero.
Best visibility margin	The margin above the threshold at the best time. A value above 0.5 indicates more comfortable visibility.
Best visibility time / Sun alt at best / Star alt at best	The time and conditions at which the star was most visible during the scan.
Twilight / Moon / Light pollution / Extinction impact	An estimate of how much each factor contributes to limiting visibility.

Moon

Column	Meaning
Moon alt / Moon illum	The Moon's altitude and illuminated fraction.
Moon-star sep	The angular distance between the star and Moon. Proximity to the Moon makes faint stars harder to see.
Moon air mass / Target air mass	The atmospheric thickness in the Moon's direction and in the star's direction.
Sky no-moon / Sky with-moon / Δ sky from Moon	The background brightness without the Moon, with the Moon, and the change caused by it.
Moon effect confidence	The confidence level of the Moon-effect estimate.
Moon affected (emp/phys)	Whether the Moon had a significant effect under the empirical or Physical model.
Limit no-moon / Δ moon limit	The limiting magnitude without the Moon and how much the Moon lowered it.

SQM Source, Probability, and Warnings

Column	Meaning
Sky baseline source	The source of baseline sky brightness: SQM, Bortle, sky-condition preset, or an SQM map.
SQM zenith input/used	The zenith SQM value entered and the value actually used.
Target SQM input/used/source	The directional SQM measurement and the source from which it was taken.
Directional SQM / interpolation / nearest	Directional SQM-map values and how the calculator interpolated among them.
Low-alt correction / Base SQM before/after	The low-altitude correction and how it changed the baseline SQM.
Visibility probability / thresholds	A verbal probability diagnostic estimated from the threshold margin. It does not replace observation.
Uncertainty reasons	Reasons for uncertainty: missing kV/SQM, a low star, Moon, haze, catalog data, and similar issues.
Sky/Catalog/Planet warning / Model limits	Warnings about sky data, catalog data, planets, and limitations of the model.

N. Practical Recommendations

Normal evening calculation

Location, date, Evening mode, 3° minimum altitude, 6.5 magnitude limit, an appropriate sky-quality setting, Moon included, and the Physical model.

Blocked horizon

Enter the horizon obstruction or raise the minimum star altitude. Do not represent a physical obstruction by changing SQM or kV.

Observing in a city

Choose Urban sky, or preferably measure SQM in the relevant direction when possible.

Haze or dust

Choose Haze / Dust. A measured kV is especially important and is preferable to the default.

Fixed stars only

Turn off the planet option.

Professional review

Open the advanced calculations, save the CSV, and always record SQM/kV, Moon, altitude cutoff, observer data, and Confidence.

O. Common Mistakes When Reading the Output

Reading First visible

It is the threshold moment, not necessarily a time of comfortable visibility. For comfortable visibility, check Best-night status and Best visibility margin.

Reading Magnitude

Do not read the numbers like an ordinary brightness scale: magnitude 1 is brighter than magnitude 5.

Entering SQM/kV without a source

A number not supported by measurement may mislead more than a general default preset.

Ignoring Confidence

Low confidence means high sensitivity to the horizon, haze, SQM, kV, and observer data.

Second-level precision

The calculated time can be mathematically precise, while real visibility still depends on transparency, haze, light pollution, and the observer.

Turning off the Moon without reason

When the Moon is above the horizon, ignoring it does not describe actual conditions; it is only a comparison scenario.

This document is a guide to using the calculator and understanding its output. It does not establish halachic conclusions and does not replace local sky measurements.